SE444: Introduction to Artificial Intelligence

Makeup Mid-Term I — Answer Sheet

Fall 2025

Date: Wednesday, December 10, 2025

Student Information

Student ID:	Student Name:

Instructions:

- Write your answers clearly in the provided spaces.
- Show all work for partial credit.
- Circle your answer(s) for multiple choice questions.

Part I. Multiple Choice (4 pts, 20 min)

Instructions: Mark **X** for correct answer(s). Each question may have 1 or more correct answers.

Option	Q1 (5m)	Q2 (5m)	Q3 (5m)	Q4 (5m)
A				
B				
C				
D				
E				

Part II. A* Search on Sliding Puzzle (6 pts, 30 min)

Context: 3-puzzle (2×2 sliding puzzle) with goal state [1, 2, 3, 0].
Initial State: $S_0 = [3, 1, 0, 2]$
Heuristic: $h(n)$ = number of misplaced tiles (excluding blank)

Question II.1: Problem Formulation (1 pt, 5 min)

Define the 6 components of the search problem formulation.

Answer:

1. **State space:**

2. **Initial state:**

3. **Actions:**

4. **Transition model:**

5. **Path cost function:**

6. **Goal test:**

Question II.2: Trace A* Search (3 pts, 15 min)

From $S_0 = [3, 1, 0, 2]$, trace A* search showing expansions, g, h, f values, and frontier updates.

Answer:

Question II.3: Comparative Analysis (2 pts, 10 min)

(a) Heuristic Comparison (1 pt):

Is Manhattan distance h' admissible? Justify:

Which heuristic expands fewer nodes? Why?

(b) Greedy vs A* (1 pt):

Is Greedy's solution necessarily optimal? Explain:

Scenario where Greedy expands FEWER nodes than A*:

Scenario where Greedy expands MORE nodes than A*:

Part III. Genetic Algorithm for Exam Scheduling (5 pts, 30 min)

Context: 4 exams (E1–E4), 3 timeslots (T1–T3), 2 halls (H1–H2).

Question III.1: Chromosome Representation (1 pt, 5 min)

(a) Chromosome for E1→T1H1, E2→T2H1, E3→T1H2, E4→T3H1:

(b) Total number of possible chromosomes (show calculation):

(c) One advantage and one disadvantage vs binary encoding:

Question III.2: Fitness Computation (2 pts, 10 min)

Compute penalties for S_1 = [T1H1, T1H2, T2H1, T2H2] and S_2 = [T1H1, T2H1, T1H1, T3H2].

Chromosome S_1 Analysis:

(a) Hard constraint violations:

(b) Soft constraint violations (student conflicts):

(c) Total penalty:

Chromosome S_2 Analysis:

(a) Hard constraint violations:

(b) Soft constraint violations (student conflicts):

(c) Total penalty:

(d) Which is fitter? Explain:

Question III.3: GA Operators and Analysis (2 pts, 15 min)

Part (a): Selection Pressure (0.5 pt)

Which strategy has higher selection pressure (tournament size 2 vs 5)?

Effect on exploration-exploitation and premature convergence:

Part (b): Two-Point Crossover (1 pt)

Parents: P1 = [T1H1, T1H2, T2H1, T2H2], P2 = [T2H1, T3H1, T1H2, T2H1]
Crossover points: after positions 1 and 3.

(i) Two offspring after crossover:

(ii) Penalty computation for one offspring:

(iii) Is the child better, worse, or in between parents? What does this illustrate?

Part (c): Mutation Analysis (0.5 pt)

Suggested mutation operator:

Problem with mutation rate too high (50%):

Problem with mutation rate too low (0.1%):

Recommended mutation rate and justification: